

Building design and operation multi-objective optimization: Energy costs vs. Emissions[☆]

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ABSTRACT

The development of new energy technologies enables the construction of sustainable energy systems, and deepens coupling between multi-energy resources and traditional energy-consuming objects, and complicates data in the design process. Meanwhile, balancing low carbon requirements and economic needs should be further optimized, and it posing a challenge for establishing an optimal energy system design method considering technical, economic, and environmental triple requirements. To address this gap, this work firstly explores a technology optimization approach for the energy supply and proposes a system holistic optimization model including two steps. The first approach utilizes Generative Adversarial Networks(GAN) to address the demand data deficiency. The second approach proposes an improved multi-objective Non-dominant Sorting Genetic Algorithm-III(NSGA-III) model to solve the non-local optimal problem of balancing two objectives. Results indicate that the proposed model achieves higher prediction efficiency than traditional methods, with an average R^2 value 13.79 % better than a baseline neural network model. Furthermore, the whole-system design approach proposed in this paper addresses the issue of different optimization ranges for various objectives, and reduces the life cycle carbon emissions (LCCE) and life cycle cost(LCC) of the optimal scheme by 34.74 % and 20.69 %, respectively, and the optimal emission is 54.78t. Multi-dimensional analysis indicates that the holistic optimization performs excellently, catering to present and future energy needs better.

1. Introduction

At present, the energy structure dominated by coal, oil and gas has caused a global energy and environmental crisis. The current energy consumption pattern is not sustainable for human society. It is imperative to carry out energy conservation and emission reduction upgrades across the entire energy system under net-zero targets, which is conducive to meeting present and future green energy needs and improving people's living standards [1,2].

The design phase presents the best opportunity for energy saving, representing the most effective measure in the overall process of energy system development. It is widely believed that the planning of multi-energy systems is a critical stage that determines energy consumption and carbon emissions performance throughout the whole system life cycle. The rationality of this stage dominates the impact of subsequent construction or operation activities on the environment and resource consumption. As an important method of architectural design

evaluation, and a development branch of techno-economic assessments (TEA) [3], life cycle assessment (LCA) focuses on assessing the overall situation of a building throughout its life cycle. In detail, minimum life cycle carbon emissions(LCCE) and life cycle cost (LCC) are commonly set as targets, which typically present a dual demand relationship throughout the life cycle, encompassing products, processes, and activities among other aspects [4,5]. LCCE focuses on evaluating carbon dioxide emissions produced throughout their entire life cycle, including all carbon equivalent emissions and outputs at different stages of the life cycle. LCC is widely used to accurately determine total costs over a specific period, and design cost usually increase with construction progress [6]. At the same time, with the deepening of multi-energy system research [7], and the rapid development of material and energy flow interaction [8], the research of combining information with life cycle assessment has attracted extensive attention [9]. Therefore, how to reasonably carry out the life cycle design of a building energy system from a whole information physical fusion system perspective has

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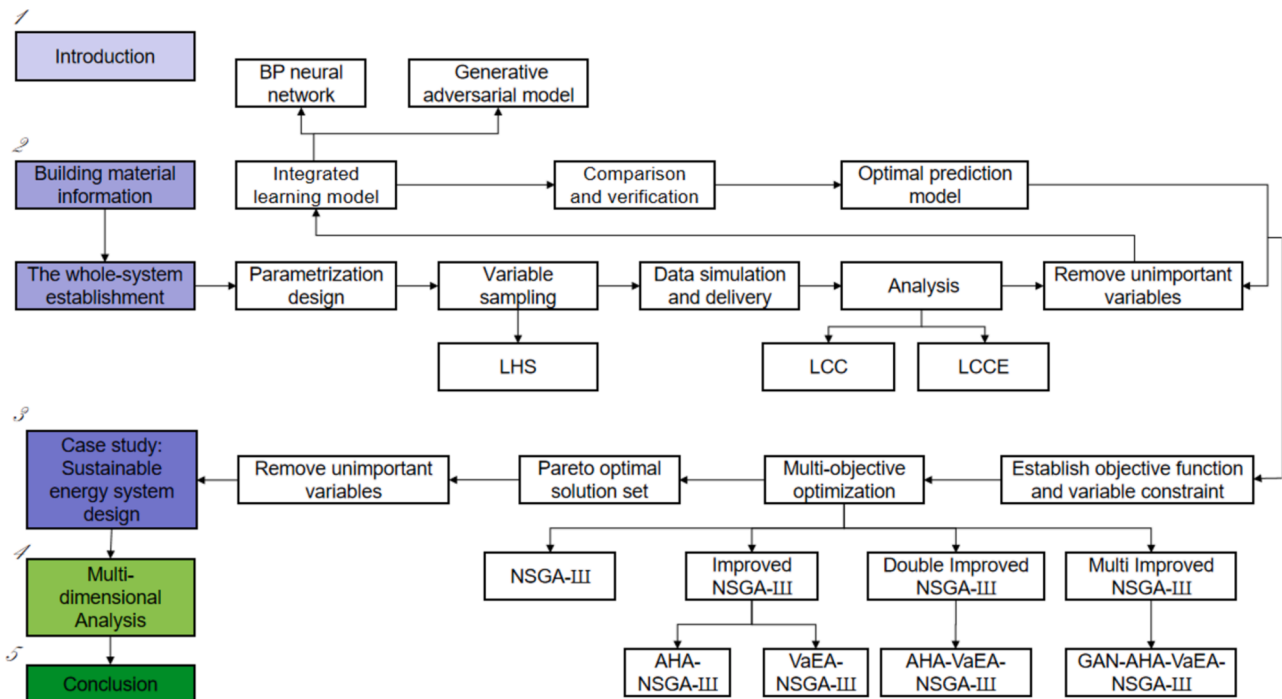


Fig. 1. The process and steps of this study.

become a new scientific problem.

To address this issue, it is essential to clarify the input mode of data processing based on energy demand prediction, and establish a reasonable model to output multi-objective optimization results. Previous studies have primarily employed two methods for energy demand setting: software modeling and data-driven approaches. Common modeling software includes EnergyPlus, eQUEST and others. However, modeling can be demanding and time-consuming. Data-driven algorithms primarily utilize machine learning and other algorithms to make predictions based on historical data. Shao et al. established an energy consumption prediction model for hotels using a support vector machine with HVAC and weather data as input parameters [10]. Tian et al. proposed a novel energy consumption predictive model called FCM-GWO-BP neural network to address the unclear physical significance, poor data generalization, and low fitting accuracy issues of the conventional backpropagation neural network model [11]. Rocio et al. identified influencing parameters including personnel density, building geometry, and natural ventilation volume [12]. Zhang et al. adopted the Clustering Decision Tree algorithm to address the issue of data imbalance and identify building operating conditions [13]. In essence, traditional methods primarily rely on model building and existing historical data. To achieve good performance, the training dataset must be representative and contain sufficient diversity. However, collecting such representative and extensive data is challenging, costly, and time-consuming [14], especially in new buildings and energy-efficient retrofitted buildings. If we lack sufficient data or if the sampled data deviates from the true data distribution, it can impact prediction accuracy.

Recent years, to overcome these problems, and enhance the learning ability of machine learning methods, and improve the quality of artificial data, and increase the prediction accuracy, researchers have turned their attention to Generative Adversarial Networks (GANs), which can generate high-quality training samples to supplement and extend real data. GAN is a cutting-edge topic in deep learning and artificial intelligence, and capable of exploring complex data distributions and generating high-quality samples that effectively supplement real data [15]. The GAN model was first proposed by Goodfellow et al in 2014 [16]. Different from traditional adversarial models, GANs do not directly estimate the probability density of each sample point in the sample

space, but instead implicitly express the distribution of the generator to the sample through the adversarial process [17]. GAN comprehensively considers numerous factors, such as changing trends and data correlations. Douzas et al. utilized GAN to approximate the real data distribution and generate data for several imbalanced datasets [18]. Literature [19] have respectively applied GAN to realize data enhancement for imbalanced data, image and sequence data to improve accuracy prediction. Esteban et al. proposed a recursive GAN and a recursive conditional GAN to generate realistic multi-dimensional medical time series [20]. Tang et al. proposed a method for programmatic data enhancement using an auxiliary classifier GAN [21]. Compared with datasets in research fields such as image recognition and speech processing, the number of datasets in building energy carbon emission and cost time series is relatively small, and there is still a lack of datasets in the field of new construction. GAN provides an effective tool for enhancing data generation and predicting building characteristics. The proposed method, which does not rely on historical data, instead making predictions based on key features has the potential to significantly contribute to research on sustainable energy system design.

Furthermore, multi-objective optimization algorithms have been widely used in architectural design optimization [22–24]. However, most genetic algorithm-based multi-objective optimization algorithms adopt simple and fixed evolutionary strategies, which are not ideal for complex applications such as architectural design optimization. The Non-dominated Sorting Genetic Algorithm-II (NSGA-II), developed in 2002, is capable of exploring the optimal solution for fast sorting selection. It maintains the optimal individual from parents in the offspring and generates a set of non-dominated Pareto optimal solutions, thereby significantly improving optimization efficiency [25]. The Non-dominated Sorting Genetic Algorithm-III (NSGA-III) is an update to the NSGA-II algorithm, utilizing a reference point mechanism to emphasize non-dominated group members and enhance the robustness and effectiveness of the results [26]. Chen et al. compared the optimization performance of NSGA-II, NSGA-III and the multi-objective evolutionary algorithm based on decomposition to determine the best building performance, and indicated that NSGA-III has the highest optimization efficiency [27]. As reported in reference [28], Zhai et al. combined the non-dominated sorting genetic algorithm(NSGA-II) with Energy Plus to

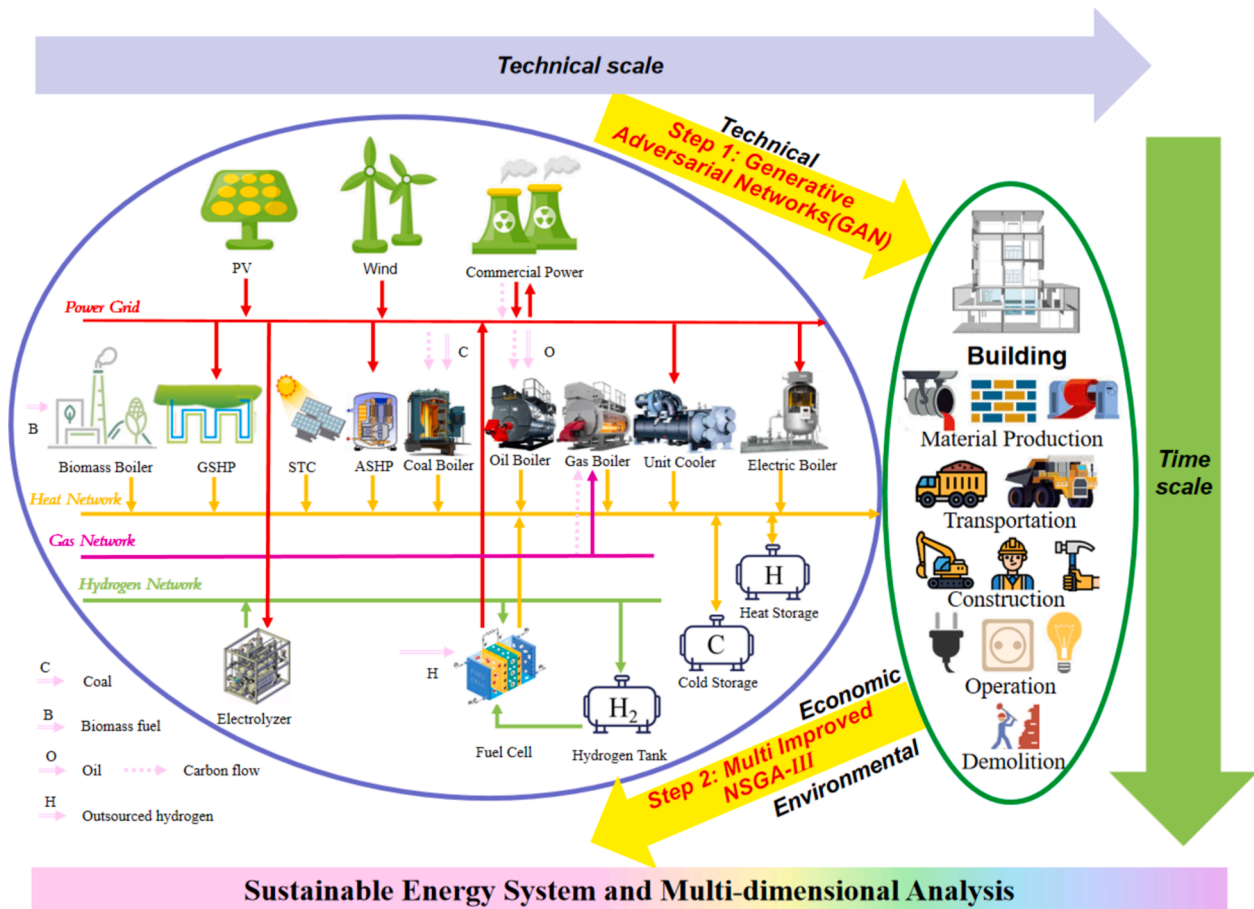


Fig. 2. The whole-system holistic optimization approach.

optimize the glass characteristics and select a design plan that balanced building energy performance. However, the aforementioned algorithm faces challenges in intensifying selection pressure on the Pareto optimal frontier and obtaining well-diversified solution sets under different optimization ranges of different objectives. Furthermore, the algorithm's accuracy and calculation speed should be improved.

Despite prior research, in the field of energy system optimization, under economic and carbon emission dual requirements, there remains a lack of efficient, convincing design methods for data deficient energy system lifetime designs. Therefore, the research takes LCCE and LCC as important optimization targets and aims to obtain a suitable design method. The proposed method of multi improved NSGA-III optimization algorithm was proposed. Analysis of total cost, renewable energy utilization, carbon emissions, and other indicators in the design stage can effectively guide the building and energy system towards uniform standards. This approach aids in determining different energy system design schemes and related parameters in a more scientific and reasonable manner.

The study integrated the energy system and building organically as a whole. The calculation and analysis of carbon emissions in the design stage can optimize system configuration and energy parameters. The energy system plan can be obtained throughout the entire life cycle via a mixed design scheme. The first part of this paper outlines the research content and the subsequent parts presenting the following: The second part primarily introduces the holistic system approach boundary to building integrated energy systems, covering model architecture, design data sources, model establishment, optimization algorithms and constraint conditions; The third part conducts a case study of an office building in Shaanxi Province, and mainly introducing feature screening, optimal design results, optimal emission reduction outputs, and other

topics; The fourth part analyzes the paradigmatic method and conducts concrete data analysis to propose an alternative optimization scheme; Finally, we summarize the contents of the article. Fig. 1 depicts the process and steps of this study.

2. The whole-system of building integrated energy system

2.1. Model architecture

This study aims to investigate integrated building energy system design across the whole-system and full life cycle, and proposes an improved multi-objective NSGA-III optimization algorithm under the whole system structure. At the physical level, the carbon emissions of the system are calculated and the main characteristics extracted. At the data level, the migration algorithm is used to train the model, yielding a high-precision model for predicting life cycle carbon emissions (LCCE) and life cycle cost (LCC). The model architecture is depicted in the figure below, and the options for energy systems including power grids, solar photovoltaic panels, cooling water storage units, hot water storage units and other low carbon energy technologies. The proposed method can effectively predict and optimize LCCE and LCC, thereby obtaining the optimal design scheme. In this study, we verify its effectiveness from the perspective of the whole life cycle performance of buildings, and add new sampling methods, algorithms and optimization algorithms to enrich its content. We propose a new method for selecting the optimal building and integrated system scheme, making the strategy more rigorous and practical. Fig. 2 depicts the whole-system holistic optimization approach. This research's originality comprises the following three elements:

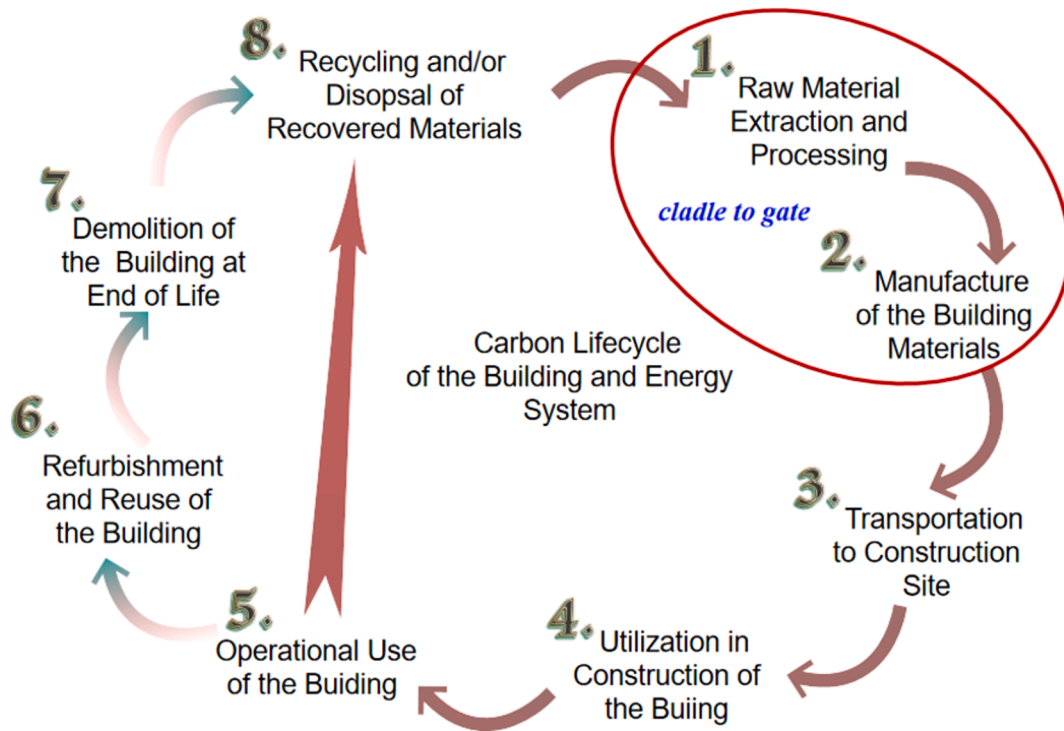


Fig. 3. The whole life cycle diagram of the building.

- A new strategy is proposed from a full life cycle perspective based on the whole system scheme, which aims to reduce LCCE and LCC and improve the feasibility of optimization results.
- The paper utilizes GANs as a data-enhancement method to facilitate feature extraction and demand prediction, while also obtaining the life cycle carbon emissions and economic performance of a building energy system.
- The multi-optimized NSGA-III method is used to reduce the influence of different target variation amplitudes on the final scheme, so that the strategy can obtain more reasonable optimization results.

2.2. Model establishment

The objective function is defined as the balanced minimum of carbon emissions and cost over the entire service life of the integrated energy system, which can be described as:

$$\min F = \min[F_e, F_c] \quad (1)$$

where F_e is the total emissions of the whole life cycle of the integrated energy system, and $F_c \setminus \text{MERGEFORMAT}$ is the cost of the integrated energy system.

2.2.1. Building life-cycle carbon emission

The core of building carbon emission calculation should be simple, consistent, user-friendly, and accurate. This study quantitatively assessed life cycle carbon emissions (LCCE) in accordance with *China's National Building Carbon Emission Standard* (GBT 51366–2019) [29]. The system's total life-cycle emissions are expressed as:

$$F_e = CE \quad (2)$$

where F_e represents the minimum carbon emission targets of the system; CE is the life-cycle carbon emission of the building body, in tons.

Measurement of building system boundaries and carbon emissions is a key part of life cycle assessment (LCA). The LCA, as defined by ISO 14040, is the process of preparing and assessing the input, output and potential environmental impact of a product system throughout its life

cycle. Before carbon emissions prediction, the study defines the target system boundaries, including building space boundaries and life cycle process boundaries. Fig. 2 illustrates the scope boundary, which corresponds to the scope of the study. It encompasses both the building body and the energy supply system. The building life cycle is divided into five main stages, referred to as *cradle to gate*, with details available in Fig. 3. These stages are: material production, transportation, construction, operation and maintenance, and demolition. The objective of the study is to develop an integrated energy system sustainability design that accounts for life-cycle carbon emissions and costs. Furthermore, the building life cycle inventory data is imported into the corresponding engineering software, Glodon.

Based on the above theory, the carbon emission of building life cycle can be expressed as:

$$F_e = CE = CE_1 + CE_2 + CE_3 + CE_4 + CE_5 \quad (3)$$

where $F_c \setminus \text{MERGEFORMAT}$ is the cost of the integrated energy system. CE is the carbon emission of the entire life cycle of the building, in tons. CE_1 , CE_2 , CE_3 , CE_4 and CE_5 are respectively the carbon emissions of the material production stage, transportation stage, construction stage, operation stage and demolition stage, in tons. The calculation method description of these can be found in the previous study [30].

The uncertainty of buildings carbon emissions is greatly affected by their use mode, including power generation and distribution, household appliances, office equipment, and cooking among others. This portion of carbon emissions does not constitute a significant proportion of the total carbon emissions, and thus does not affect the judgement of the carbon emission intensity of the building scheme in the design stage. International practice is that building carbon emissions calculations do not include household appliances, office appliances, or cooking carbon emissions. Our research is conducted in compliance with this requirement.

2.2.2. Building life cycle cost

The building life cycle benefit is significant to owners. Numerous methods exist for evaluating the life cycle cost (LCC) of buildings,

including net present value and annuity. In this study, LCC is calculated according to the **EN 15459** definition, which assesses the life cycle economics of a project based on net present value [31]. By estimating the initial investment cost, annual net operation and maintenance cost, final disposal cost and salvage value, the LCC of each measure was calculated. The available integrated energy system cost is expressed as:

$$F_c = \sum_{n=1}^{14} CRF \cdot M \cdot C_n^{\text{invest}} + M \cdot C_{O\&M} - C_{re} - \zeta \quad (4)$$

where CRF is the capital recovery coefficient. M is the system service life in years. C_{invest} is the initial investment cost in RMB. $C_{O\&M}$ is the operation and maintenance cost, in RMB. C_{re} is the equipment salvage value in RMB. And ζ is the predetermined renewable energy subsidy, in RMB. The research considers 14 energy supply technologies including PV, wind and so on, as presented in Fig. 2.

The capital recovery coefficient is:

$$CRF = \frac{i(1+i)^m}{(1+i)^m - 1} \quad (5)$$

where i is the annual interest rate, which is generally 2 %; and m is the useful life of the system, which is 15 years.

2.2.3. Constraints and decision variables

To further explain the number of variables or degrees of freedom involved in the research and better introduce the optimization problem,

the following subsystem constraints are listed:

The energy balance limitations of electricity, heating and cooling

$$P_g^k + P_{pv}^k + P_{wtg}^k + P_{hy,out}^k = P_d^k + P_{ashp}^k + P_{eb}^k + P_{gshp}^k + P_{hy,in}^k + P_{ru}^k, \forall k \in \{1, 2, \dots, K\} \quad (6)$$

where P_d^k represents the electricity, heat and cold energy demands of buildings. P_g^k represents the grid generation. P_{pv}^k represents the photovoltaic power generation. P_{wtg}^k represents the wind power generation. $P_{hy,out}^k$ represents the hydrogen subsystem power generation. P_{ashp}^k represents the air source heat pump electricity demand. P_{eb}^k represents the electricity demand for electric boilers. P_{gshp}^k represents the ground source heat pump electricity demand. $P_{hy,in}^k$ represents electricity demand of hydrogen energy subsystem. P_{ru}^k represents electricity demand for chillers. K represents the time period.

$$G_{ashp}^k + G_{stc}^k + G_{bio,out}^k + G_{eb}^k + G_{hy,out}^k + G_{cfb}^k + G_{gfb}^k + G_{ofb}^k + G_{gshp}^k + G_{wt,out}^k = G_d^k + G_w^k + G_{wt,in}^k + G_{wt,loss}^k, \forall k \in \{1, 2, \dots, K\} \quad (7)$$

where G_d^k represents the heat energy demands of buildings. G_{ashp}^k Ground source heat pump heating. G_{stc}^k represents the solar collector heating. $G_{bio,out}^k$ represents the biomass boiler heating. G_{eb}^k represents the electric boiler heating. $G_{hy,out}^k$ represents the hydrogen energy subsystem

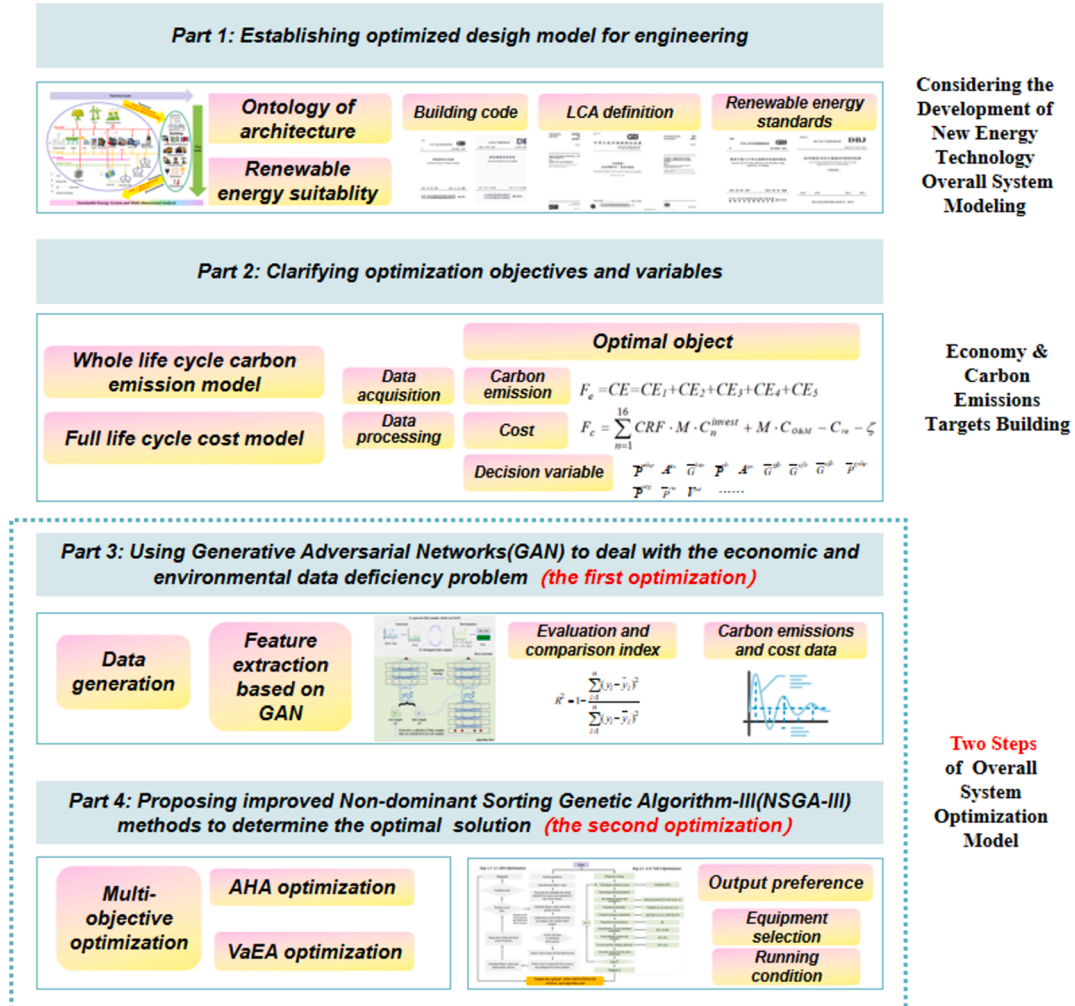


Fig. 4. Optimal design framework.

heating. G_{cb}^k represents the coal boiler heating. G_{gb}^k represents the gas boiler heating. G_{ob}^k represents the oil-fired boiler heating. G_{gshp}^k represents the air source heat pump heating. $G_{wt,out}^k$ represents the hot water tank heating. G_w^k represents the domestic hot water demand. $G_{wt,in}^k$ represents the hot water tank input. $G_{wt,loss}^k$ represents the hot water tank loss. K represents the time period.

$$Q_{ashp}^k + Q_{gshp}^k + Q_{ct,out}^k + Q_{ru}^k = Q_d^k + Q_{ct,in}^k + Q_{ct,loss}^k, \forall k \in \{1, 2, \dots, K\} \quad (8)$$

where Q_d^k represents the cold energy demands of buildings. Q_{ashp}^k represents the air source heat pump cooling. Q_{gshp}^k represents the ground source heat pump cooling. $Q_{ct,out}^k$ represents the chiller output. Q_{ru}^k represents the chiller for cooling. $Q_{ct,in}^k$ represents the cold water tank demand. $Q_{ct,loss}^k$ represents the cold water tank loss. K represents the time period.

Hydrogen energy subsystem

$$G_{hy,out}^k = \beta_{fc,g} m_{hy,out}^k, \quad (9)$$

$$m_{hy}^{k+1} = m_{hy}^k + m_{hy,pur,in}^k + \beta_{el+co} P_{hy,in}^k - P_{hy,out}^k / \beta_{fc,e}, \quad (10)$$

$$P_{hy,out}^k / \tau \leq \bar{P}_{fc}, P_{hy,in}^k / \tau \leq \bar{P}_{el}, m_{hy}^k \leq M_{hy}, \quad (11)$$

where $\beta_{fc,g}$ represents conversion efficiency. $m_{hy,out}^k$ represents the mass of electrolytic hydrogen over period k , in time t . $m_{hy,pur,in}^k$ represents the amount of hydrogen bought over period k , in time t . β_{el+co} represents the efficiency of electrolytic cell and conversion. m_{hy}^k represents the amount of hydrogen over period k . $\beta_{fc,e}$ represents conversion efficiency, and τ represents time. K represents the time period.

(3) Ground source heat pump and the corresponding soil heat storage system unit

$$CE_{r,gshp} = \frac{m_{r,gshp}}{y_{gshp}} GWP_{r,gshp} / 1000, \quad (12)$$

$$G_{gshp}^k = \eta_{gshp,g} P_{gshp,g}^k \cdot \tau, \quad (13)$$

$$Q_{gshp}^k = \eta_{gshp,q} P_{gshp,q}^k \cdot \tau, \quad (14)$$

$$(P_{gshp,g}^k + P_{gshp,q}^k) / \tau \leq \bar{P}_{gshp}, \quad (15)$$

where $m_{r,gshp}$ represents the refrigerant quality of the ground source heat pump in kg. y_{gshp} represents the ground source heat pump life in years. $GWP_{r,gshp}$ represents the global warming potential. $GWP_{r,gshp}$ represents the heating of the ground source heat pump over period k , in kW. Q_{gshp}^k represents the cooling of the ground source heat pump over period k , in kW. $\eta_{gshp,g}$ represents the heating conversion efficiency of the ground source heat pump. $\eta_{gshp,q}$ represents the cooling conversion efficiency of the ground source heat pump. P_{gshp}^k represents the total power input to the ground source heat pump heating over period k , in kW. $P_{gshp,q}^k$ represents the power input to the ground source heat pump cooling over period k , in kW. $P_{gshp,g}^k$ represents the heating input electricity of the soil heat storage system over period k , in kW. τ represents time.

(4) Air source heat pump

$$CE_{r,ashp} = \frac{m_{r,ashp}}{y_{ashp}} GWP_{r,ashp} / 1000, \quad (16)$$

$$G_{ashp}^k = \eta_{ashp,g} P_{ashp,g}^k \cdot \tau, P_{ashp,q}^k \leq \bar{P}_{ashp}, \quad (17)$$

$$Q_{ashp}^k = \eta_{ashp,q} \cdot P_{ashp,q}^k \cdot \tau, P_{ashp,q}^k \leq \bar{P}_{ashp}, \quad (18)$$

where $m_{r,ashp}$ represents the refrigerant quality of an air source heat pump, in kg. y_{ashp} represents the lifespan of the air source heat pump, in years. $GWP_{r,ashp}$ represents the global warming potential. G_{ashp}^k represents the heating over period k , in kW. Q_{ashp}^k represents the cooling over k period, in kW. η is conversion efficiency.

In addition, it encompasses an electric boiler, photovoltaic, solar collector, wind power generation, cold water tank, and hot water tank model, along with other basic equipment models not described here but available in the literature [32].

The optimal design framework is depicted in Fig. 4. The proposed mixed integer programming problem, which incorporates two objectives of low-carbon economy, and can be utilized in conjunction with the multi improved NSGA-III algorithm to obtain the corresponding optimal target point. Based on the optimal target point, the carbon emission decision optimization scheme for the comprehensive energy system can be derived, and the final decision parameters or optimization variables are expressed as:

$$X^d = [P^{cp}, \bar{P}^{ashp}, A^{stc}, \bar{G}^{bio}, \bar{P}^{eb}, A^{pv}, \bar{G}^{gfb}, \bar{G}^{cfb}, \bar{G}^{ofb}, \bar{P}^{gshp}, \bar{P}^{wtg}, \bar{P}^{ru}, \bar{P}^{fc}, V^{wt}, V^{ct}, CCER]$$

where P^{cp} represents the power input from grid. $\bar{P}^{ashp}$ represents the rated power of the air source heat pump equipment obtained by selection. A^{stc} represents the area of the solar collector obtained by selection. $\bar{G}^{bio}$ represents the rated heat of the biomass boiler equipment obtained by selection. $\bar{P}^{eb}$ represents the rated input power of the electric boiler equipment obtained by selection. A^{pv} represents the area of the photovoltaic equipment obtained by selection. $\bar{G}^{gfb}$ represents the rated heat of the gas boiler equipment obtained by selection. $\bar{G}^{cfb}$ represents the rated heat of the coal-fired boiler equipment obtained by type selection. $\bar{G}^{ofb}$ represents the rated heat of the oil-fired boiler equipment obtained by type selection. $\bar{P}^{gshp}$ represents the rated heat of the ground source heat pump equipment obtained by type selection. $\bar{P}^{wtg}$ represents the rated input power of the wind turbine equipment obtained by type selection. $\bar{P}^{ru}$ represents the rated input power of the unit cooler obtained by type selection. $\bar{P}^{fc}$ represents the rated input power of the fuel cell obtained by type selection. V^{wt} represents the hot water tank capacity obtained by type selection. V^{ct} represents the cold water tank capacity obtained by type selection. And CCER represents the optimal certified carbon emission reduction.

2.3. Basic data acquisition

Primary data can be obtained by reviewing design drawings, purchase lists and other engineering construction related technical data. It is straightforward to ascertain the consumption of various building materials required for construction projects through the bill of quantities, the bill of materials and so on.

During project exploration, the geological exploration rig consumes energy. However, considering its short working time and low energy consumption, the calculation time boundary for carbon emission in the construction stage is stipulated to start from the tool's arrival on site.

In the production of building materials, if low-value waste is used as a raw material, the carbon footprint of its upstream process can be overlooked. If other recycled materials are used, they should be calculated as 50 % of the carbon emissions of the primary materials they replace. Renewable construction waste generated during the construction and demolition phases of a building, accounts for 50 % of the carbon emissions associated with primary materials. This waste should be properly managed and replaced with more sustainable alternatives, in

Table 1
Representative design parameter and their source.

Restrictions	Source
The design life is typically chosen to be 50 years, with other options available, ect.	Building Carbon Emission Calculation Standard (GB/T 51366-2019) [29]
The total window wall area ratio of the building should not exceed 0.50, ect.	General Code for Building Energy Efficiency and Renewable Energy Utilization (GB 55015-2021) [32]
The reflectance of the floor is 0.3, wall is 0.6, outer surface is 0.5, and ceiling is 0.75, ect.	Evaluation Standard for Green Building (GB/T 50378-2019) [33]
The relative humidity in the room generally fall within the range of 20 % to 80 %, ect.	Code for Thermal Design of Civil Buildings (GB 50176-2016) [34]

order to reduce the overall carbon footprint of the building.

During building construction, the energy power required for the operation of construction machinery and small equipment is a major source of carbon emissions. Normal respiration and carbon dioxide release are normal physiological processes for humans, and there is no significant difference in the amount of carbon dioxide breathed and released by construction workers compared to the general population, so it is not included in the carbon emissions of construction workers. Materials, components and parts used in construction, such as ready-mixed concrete, concrete components, precast piles, doors and windows, are typically produced off-site and therefore excluded from energy consumption during the construction phase. The energy consumption of materials, components and parts mixed and produced on-site should be

included. To facilitate the turnover of the structure, the construction phase typically utilizes sandwich color steel plate for temporary structures, such as movable boards, houses and container houses. These simple temporary houses are easy to install and dismantle, requiring little energy for construction and dismantlement, and are not included in carbon emissions calculations during the building's construction phase.

The design constraints of the building energy system must meet relevant policies and regulations. Specifically, the constraint data of the experimental design were sorted out as shown in the Table 1 below. Other relevant national and local standards, including codes and regulations, national standards, current design and construction codes, and relevant laws and regulations, as well as rules and guidance documents, procedures, and energy-saving regulations of the province, among others, are not detailed due to the length limit.

Based on the above description, experiments and analyses will be conducted. Technical costs, efficiency data, energy market conditions and climate conditions align with national conditions. The *Chinese Meteorological Data Set* provides outdoor parameters. By combining meteorological parameters from the *Design Code for Heating, Ventilation and Air Conditioning of Civil Buildings (GB50736-2012)* with hourly and daily variations of outdoor dry bulb temperature in typical years we extracted data to obtain results for each day. Simultaneously, this paper collects experimental parameters through actual research, and sets the price of hydrogen at 1.6 yuan/standard cubic meter, or 30 yuan/kg; and calculates the system life as 15 years. The photoelectric energy conversion efficiency is 16 %. The efficiency of hydrogen-electric energy conversion is 45 %. And the hydrogen-electric heating energy conversion efficiency is 90 %. Specific parameters can be obtained during model establishment.

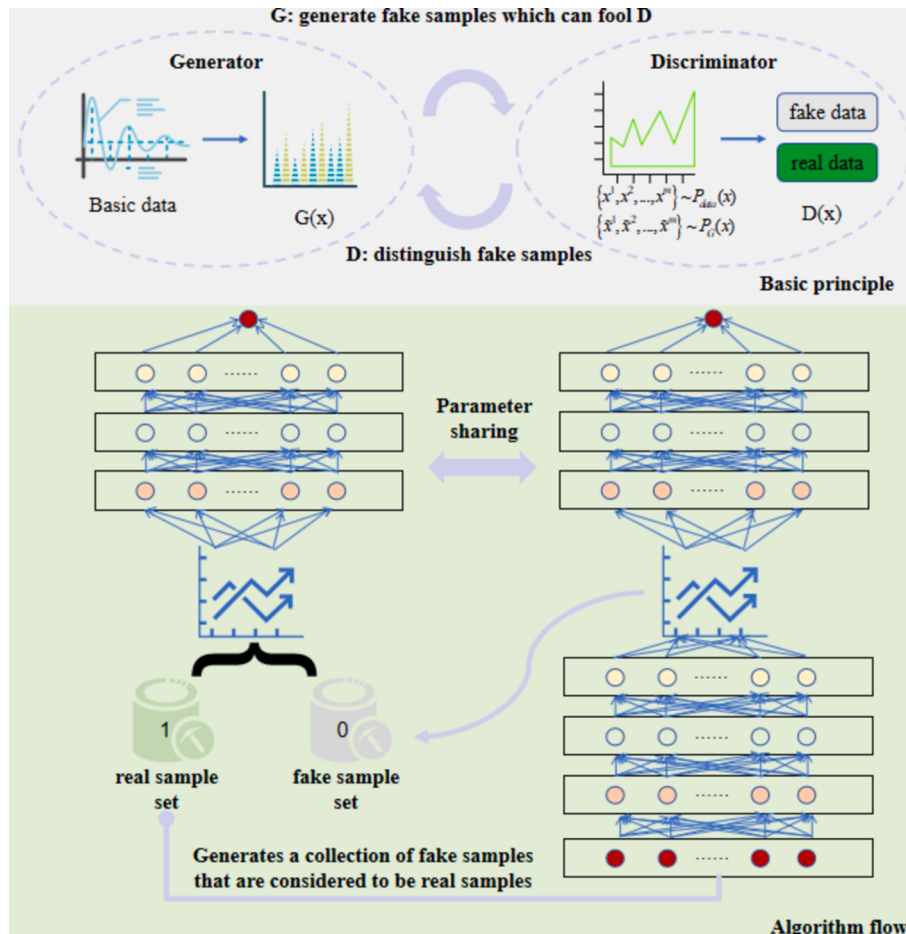


Fig. 5. Generative Adversarial Networks basic principle and algorithm flow.

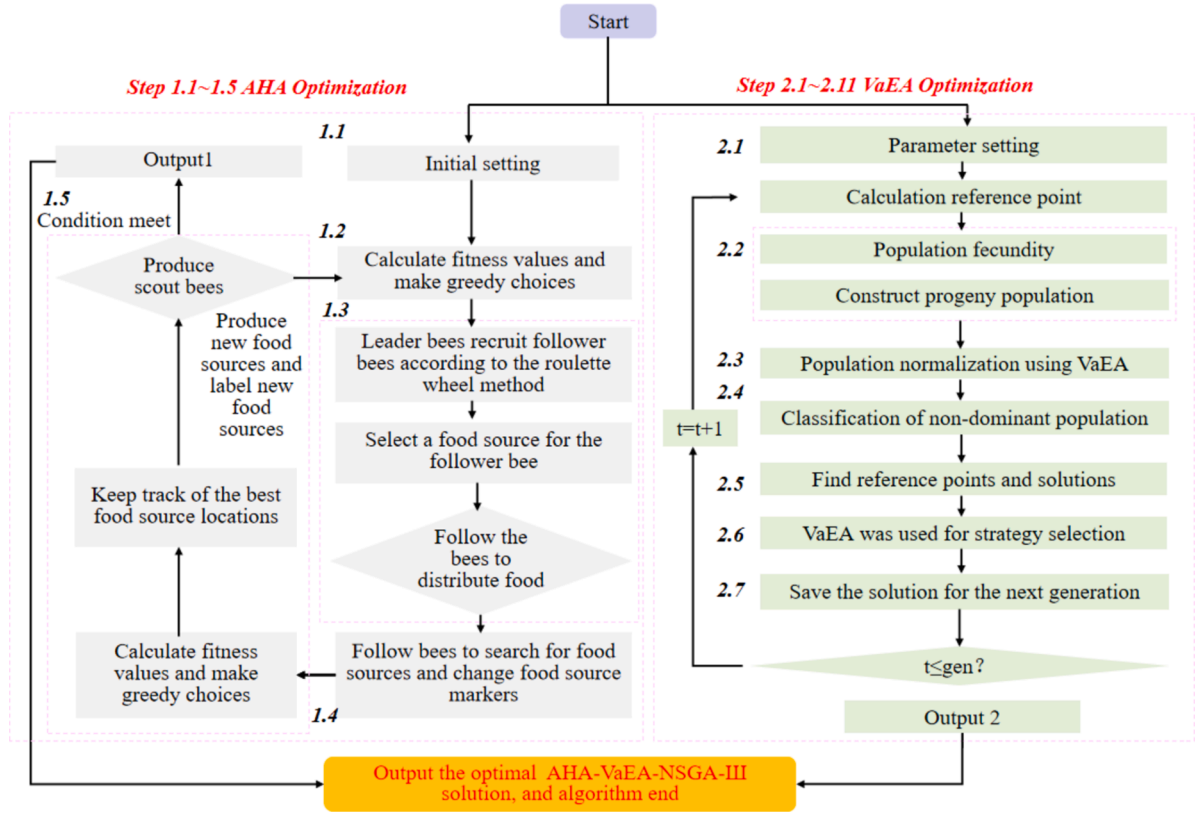


Fig. 6. Algorithm flow of AHA-VaEA-NSGA-III.

2.4. Data processing

The dataset was randomly divided into 30 % test data and 70 % training data before training the machine learning model on the training data. Equation (19) was utilized to normalize the data in this study because input parameters with varying units and physical meanings [35]. A larger positive index value indicated better system performance, and this value is positively correlated with the evaluation result.

$$y_{ij} = \frac{x_{ij} - \min_{1 \leq i \leq m} x_{ij}}{\max_{1 \leq i \leq m} x_{ij} - \min_{1 \leq i \leq m} x_{ij}} \quad i = 1, 2, \dots, m; j = 1, 2, \dots, n \quad (19)$$

where $\min_{1 \leq i \leq m} x_{ij}$ and $\max_{1 \leq i \leq m} x_{ij}$ means the max value and min vale of the index respectively.

After the above steps in the same direction and standardization, all input and output indicators can be converted into positive indicators, and the value of indicators is between [0,1].

2.5. Prediction and evaluation

2.5.1. Prediction model

GAN proposed by Ian Goodfellow et al. in 2014, and published in the paper **Generative Adversarial Networks**, which is mainly inspired by the zero-sum game idea in game theory. As shown in Fig. 5, in deep learning neural networks, G (Generator) and D (Discriminator) are in a constant game-like interaction where they attempt to outsmart each other in order for G to learn the distribution of the data. The relationship is as follows: the discriminator punishes the generator, benefiting itself while the generator loses. The generator evolves such that the discriminator punishes itself less, benefiting the generator and causing the discriminator to lose. In the GAN framework, the generator and discriminator models engage in continuous training game with each other, aiming to bring the generator samples closer to the real data

distribution, and improve the discriminator's ability to distinguish between generator samples and real data. During training, the goal of the generator is to deceive the discriminator, while the goal of the discriminator is to distinguish between samples generated by the generator and real data. Random search and grid search methods are used to select appropriate hyperparameters in the neural network to realize the operation of the prediction model [36].

In detail, G is a generative network that receives random noise. D is a discriminant network that determines whether a given set of data is real; its input parameter is x. The output $D(X)$ represents the probability that X is the real data. The output of 1 indicates that the data is 100 % genuine, while an output of 0 suggests that it is impossible for the data to be authentic. The algorithm steps are as follows:

Step 1. Initialize θ_d for D and θ_g for G.

Step 2. Learning D through repeating k times and finding lower found of $\max_D V(G, D)$.

- (1) Sample m examples $\{x^1, x^2, \dots, x^m\}$ from data distribution $P_{data}(x)$.
- (2) Sample m noise samples $\{z^1, z^2, \dots, z^m\}$ from the prior $P_{prior}(z)$.
- (3) Obtaining generated data $\{\tilde{x}^1, \tilde{x}^2, \dots, \tilde{x}^m\}, \tilde{x}^i = G(z^i)$.
- (4) Update discriminator parameters θ_d to maximize as below.

$$\tilde{V} = \frac{1}{m} \sum_{i=1}^m \log D(x^i) + \frac{1}{m} \sum_{i=1}^m \log(1 - D(\tilde{x}^i)) \quad (20)$$

$$\theta_d \leftarrow \theta_d + \eta \nabla \tilde{V}(\theta_d) \quad (21)$$

Step 3. Learning G only once.

- (1) Sample another m noise samples $\{z^1, z^2, \dots, z^m\}$ from the prior $P_{prior}(z)$.
- (2) Update generator parameters θ_g to minimize as below.

$$\tilde{V} = \frac{1}{m} \sum_{i=1}^m \log D(x^i) + \frac{1}{m} \sum_{i=1}^m \log(1 - D(G(z^i))) \quad (22)$$

$$\theta_g \leftarrow \theta_g - \eta \nabla \tilde{V}(\theta_g) \quad (23)$$

Finally, the value of V can be calculated as follow:

$$V = E_{x \sim P_{data}} [\log(D(x))] + E_{x \sim P_G} [\log(1 - D(x))] \quad (24)$$

2.5.2. Evaluation and comparison indicators

The mean absolute error (MAE), root mean square error (RMSE), and coefficient of determination (R^2) are commonly used evaluation metrics for assessing prediction accuracy [37]. Considering consistency, this study selects R^2 for comparison. The lower the R^2 , and the greater the difference between the two sets of data [38]. The calculation formula is shown in the equation below.

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (25)$$

where y_i represent the actual value. And $\hat{y}_i$ is the predict value, while n is the total number.

2.6. Multi-objective optimization

The constructed multi-objective model has different physical meanings for the objects and a large difference in orders of magnitude, which requires careful consideration. Considering the sequential characteristics of the system, this study adopted the normalization distance between the objective function values of each group of non-inferior solutions in the Pareto frontier and the ideal point as the evaluation index. Then, select a group of non-inferior solutions closest to the ideal point as the best operation scheme. Here, the ideal point refers to the point on the Pareto frontier that minimizes the objective function, which is the origin of the coordinate axis after normalization. The distance refers to the spatial distance between two points in the DeKale coordinate system. Based on NSGA-III [39], a double-improved algorithm named AHA-VaEA-NSGA-III was proposed to further enhance the robustness and speed of the model. To ensure convergence of the solution set, the model selects the solution with best convergence as the non-dominant solution in the same small scale. To maintain diversity in the solution set, in any two different small territories, the solution with good convergence and distribution in one small territory dominates the solution with poor convergence and distribution in the other small territory. The proposed domination relation is embedded in VaEA to replace the Pareto domination relation. To reduce computational burden and improve model accuracy, the Artificial Hummingbird Algorithm (AHA) is utilized to optimize the model. Fig. 6 depicts an algorithm consisting of three stages, whose details are described as follows.

The first stage is the Artificial Hummingbird Algorithm.

Step 1.1. In the search space, the position of the t^{th} hummingbird in the population was selected, and the preset upper and lower bounds of variables were satisfied.

Step 1.2. Hummingbirds constantly seek out better food sources and exhibit superior adaptation values.

Step 1.3. During the search process, the best individual is defined as the territorial bird, while the rest are considered follower birds. Next, when there is no approaching bird from the territorial bird, the heuristic algorithm is used to determine the positions of the follower birds after t iterations.

Step 1.4. When a territorial bird detects a potential intruder, it will issue a warning. Upon receiving the warning, the follower bird will fly to the surrounding area. If the choice is better, j follower birds will fly towards it; otherwise, they will stay away.

Step 1.5. Control the search boundary and iterate until convergence to obtain the initial operation strategy of the system.

The second stage is the maximum vector angle optimization strategy based on VaEA, which improves the direction of the search evolutionary population.

Step 2.1. Define a set of reference points and initialize the total number of iterations N, which represents the maximum allowed number of iterations. The mutation rate and crossover rate, denoted by P_i and Q_i respectively, are determined by the population in the t^{th} generation. The initial population of the next generation, consisting of N individuals is selected from $R_i(P_i \cup Q_i)$.

Step 2.2. Sort R_i using non-dominant sorting into several different non-dominant layers denoted as S_i , and uniformly distributed reference points are internally and externally generated.

Step 2.3. Apply the VaEA simple normalization method to determine the ideal point and the lowest point. Based on the minimum and maximum target values in the current population, the target values of the j^{th} generation as standardized.

Step 2.4. Assuming F_l is the last non-dominant layer and its population size S_l is greater than N for the first time, a reference point is used to find the optimal number of remaining P_{l+1} individuals in F_l , and the next generation population P_{l+1} is added.

Step 2.5. Standardize all individual objective value points in reference such that both sets fall within the same range. And correlate the reference points of each individual in S_l , and obtain the remaining individuals for S_l from layer l.

Step 2.6. In each iteration, select the reference point j^* with the least number of habitats of VaEA. When the number is determined to be 0, the individual with the minimum vertical distance from the reference line is selected from Ij^* . When the number is greater than 0, an individual is randomly selected from Ij^* . For the random process that selects individuals to enter P_{t+1} when their value is greater than 0, we utilize a vector angle selection strategy based on VaEA to optimize the process. The individual with the maximum vector angle and the individual with the minimum vector angle in F_l are denoted as P_{t+1} .

Step 2.7. If the number of optimization of generation offspring population is less than or equal to the maximum number threshold, return to program start; otherwise, the operation stops, S_t becomes a new population, and the second initialization operation strategy of the system is obtained.

The third stage improved non-dominated sorting genetic algorithm is used: The first initial operation strategy and the second initial operation strategy are weighted to obtain the final operation decision of the system. The optimization of the whole system is a real-time tracking, constantly updating, and constantly correcting dynamic process.

After normalization, the distance between the Pareto frontier and the origin is calculated. The dynamic optimization algorithm NSGA-III is used to randomly select the reference set Z' , and the initial population P_i with a quantity of N is randomly generated within the reference set for random selection. Crossover of two points is simulated to achieve polynomial variation, producing the subgeneration population Q_i with quantity N. It is combined with the initial population to get a new population R_i , which is further sorted and the output of the optimal solution is realized in the case of maximum iteration. This algorithm uses a reference point based method to select individual offspring. The improvement of selection methods can address the problem of insufficient convergence and diversity in multi-objective optimization, making it easier to find the global optimal solution.

3. Case study

3.1. Research object

The case study focuses on an office building in Shaanxi Province. The total solar radiation in the area is greatest in summer, least in winter, and mid-range in spring and autumn. The total solar radiation in the area is 6.13×10^{13} kW·h, and the usable amount of solar energy resources can reach 1226.49 kW·h. The annual total solar radiation in the

Table 2

Introduction of research object initial design parameter.

Category	Data
Location	Shaanxi, China
Building area	4207.93 m ²
Floor space	2806.26 m ²
Height of building	9.33 m
Type	office building
Shape factor	0.52
Floors	4.00
Climate zone	cold climate area
Service life	50 years of building, and 15 years of energy system

Note: Floor space represents the maximum floor area of the building. Building area represents the total area of the building floors.

area is between 4970.34 to 5163.29 MJ/m², and the solar energy resources are very stable with high potential utilization value. At the same time, assuming the system has access to the power grid, gas grid and district heating network, gas boilers can also be installed in the low-carbon building pilot project. Therefore, it is possible to study the optimal amount of carbon reduction in the whole life cycle of buildings without additional energy constraints.

The floor space of the case is 2806.26 m². The total building area of the case is 4207.93 m². Table 2 presents the construction details for the case. The equipment schedule, lighting and personnel activities in this study were developed according to the actual situation of the case building. In terms of area load, according to ASHRAE standard, the room penetration rate is 0.0003 m³/s/m² and the area ventilation rate is 0.00165 m³/s/m² facade when the building is in normal operation. The building staff to space ratio is set to 0.15 persons per square meter. The heating setpoint for indoor air temperature is 22.50, and the lighting schedule is from 7:00 am to 12:00 am. Daylight performance is year-round, from 7:00 am to 6:00 pm. The above default settings comply with ASHRAE requirements and take into account local requirements and norms for activity levels and lighting density. Since the case houses utilize central air conditioning, the same HVAC system is used in the model. Based on the input parameters, the heating set point of the air conditioner is 18.21 °C and the cooling set point is 23.53 °C. This means

that if the temperature is below 18.21 °C or above 23.53 °C, the air conditioning in the office building will operate, consistent with the actual situation in the case building. The U value of the wall, roof, and window also aligns with actual building code requirements [32–34].

This study first addresses the issue of universal building energy demand acquisition. The input parameters are normalized, and reference values are established based on clear maximum and minimum values. Table 3 presents the optimization features and their variation range. A total of 28 input features are utilized, categorized into internal changes (1–4), building envelope (5–17), and system operation (18–28). Variables are labeled in bold, while non-variables are inputted. The model constraints are primarily chosen based on common research features of LCCE and LCC of the building [40–42].

Constraint parameters are a technique for constraining conditions such as the type, range, and input parameters of a function or method. Before the experiment, we restricted the range of each input parameter, as shown in the following table. The data in the table are non-design variables, which are used to limit the value range of input parameters, improve code readability, enhance matching with facts, and avoid errors from parameter mismatch. The parameter is a floating-point number, retaining two decimal places. The reference item provides an example of calculating the maximum and minimum value of the reference value.

Among them, the window-to-wall ratio refers to the ratio of the total area of external windows (including transparent curtain walls) in a given direction to the total wall area (including window areas) in the same direction. The form factor refers to the ratio of the external surface area of a building to its volume. The envelope structure heat transfer coefficient indicates the resistance of building external walls, roofs, windows, and other envelope materials to heat conduction. Relevant equipment parameters can be obtained from the Standard Code for Thermal Design of Civil Building (GB 50176-2016) [34].

Furthermore, as analyzed in section 2.3, there are multiple sources of carbon emission factors, and the same coefficient from different sources may have conflicting values. The study's database is based in China, and the coefficient prioritization principle is as follows: Firstly, data collected should be as up-to-date as possible, favoring national data over international data. Secondly, for carbon emission sources with different

Table 3

Presents the optimization features and their range of variation (The first step: input parameters of prediction algorithm).

Number	Parameter	Units	Max	Min	Reference Value	Reference 0 ~ 1 Standardization
1	Outdoor temperature	°C	39.00	−33.00	8.50	0.58
2	Relative humidity	%	80.00 %	20.00 %	43.60 %	0.39
3	Indoor temperature	°C	26.00	20.00	24.00	0.67
4	CO₂ concentration	ppm	1000.00	500.00	700.00	0.40
5	Window-wall ratio(WWR)	/	0.90	0.00	0.40	0.44
6	Building area	m ²	4207.93	19.28	4207.93	1.00
7	Wall U value	W/(m ² ·K)	0.80	0.14	0.33	0.29
8	Wall heat absorption	W	0.90	0.10	0.90	1.00
9	Wall solar energy absorption	W	0.90	0.10	0.75	0.81
10	Wall density	kg/m ³	1250.00	400.00	800.00	0.47
11	U value of roof	W/(m ² ·K)	2.00	0.16	0.42	0.14
12	Roof heat absorption	W	0.90	0.10	0.75	0.81
13	Rooftop solar absorption	W	0.90	0.10	0.70	0.75
14	Roof materials density	kg/m ³	1650.00	700.00	1500.00	0.84
15	U value of window	W/(m ² ·K)	6.00	1.00	2.50	0.30
16	Window solar heat absorption coefficient	W	0.90	0.10	0.70	0.75
17	Building orientation	/	360.00	0.00	0.00	0.00
18	Supply chilled water temperature	°C	15.00	5.00	10.00	0.50
19	Chilled water return temperature	°C	27.00	12.00	19.50	0.50
20	Cooling water supply temperature	°C	33.00	30.00	32.00	0.67
21	Cooling water return temperature	°C	39.00	35.00	38.00	0.75
22	Fresh air volume	m ³ /s/person	0.03	0.00	0.008	0.27
23	Heating temperature in winter	°C	22.50	18.00	20.00	0.44
24	Coil heating efficiency	/	0.88	0.72	0.80	0.50
25	Coil cooling COP	/	4.50	3.00	3.50	0.33
26	Cooling temperature in summer	°C	28.00	23.50	26.00	0.17
27	Number of energy users	person	72.00	0.00	12.00	0.17
28	Illumination intensity	lux	7.00	2.00	5.00	0.60

Table 4

Core parameter table of energy system optimization equipment (The second step: multi-objective optimization of energy system).

Classes	Parameter	Number	Input	Output
PV	Photovoltaic efficiency	0.16	solar power	electricity
Wind	Wind energy conversion efficiency	0.39	wind	electricity
Commercial power	Electricity price	0.58 CNY/kWh	coal	thermal
Biomass boiler	Biomass fuel price	0.85 CNY/kg	biomass	thermal
Ground source heat pump	COP	4.00	geothermal	thermal
Solar thermal collector	Produces heat per unit area	3.2KW/m ²	solar power	thermal
Air source heat pump	COP	5.00	air	thermal
Oil boiler	Efficiency	0.92	oil	thermal
Gas boiler	Efficiency	0.90	gas	thermal
Coal boiler	Efficiency	0.88	coal	thermal
Electric boiler	Efficiency	0.97	electricity	thermal
Water chilling unit	COP	3.95	electricity	thermal
Hydrogen system	Fuel cell conversion coefficient	15 kWh/kg	hydrogen, electricity	thermal and electricity
Energy storage device	Loss ratio	1.00 %	/	thermal and hydrogen

Note: COP stands for energy efficiency ratio, and each type of electricity/heat has at least one energy device to meet the energy demand.

coefficients across different scales or categories, general values rather than specific values are chosen. Thirdly, while most carbon emission coefficients can be extracted directly from existing carbon emission related databases and publications, indirect calculations can be considered for factors that cannot be directly quantified. If the ratio of carbon emission data from the production and construction of main building materials to the complete data exceeds 95 %, the calculation is considered feasible.

The data in the table shows that air source heat pumps and ground source heat pumps do not need to be installed simultaneously, and that fuel cells can provide both electricity and heat. Similarly, coal-fired boilers, oil-fired boilers, and natural gas boilers can be used as heating options, but do not need to be used simultaneously. Therefore, the scenarios that can be selected for building energy system optimization include electric cooling and heating modes, conventional electric cooling and heating modes plus photovoltaic generation, etc. The multi-

objective optimization problem is then solved. The 14 equipment energy supply types considered in the model, such as solar energy, are shown in Fig. 1, and a portion of the parameters are shown in Table 4. In this step, the objective function is to minimize carbon emissions and costs, and the optimization variable is equipment selection and capacity. In this study, we primarily compare overall optimization algorithms for input and output. The input parameters, or non-design variables, in the optimization stage are shown in the table. Equipment with the same energy output can be replaced, but the electrical and thermal demands of the building must be met.

The data in the table shows that air source heat pumps and ground source heat pumps do not need to be installed simultaneously, and that fuel cells can provide both electricity and heat. Similarly, coal-fired boilers, oil-fired boilers, and natural gas boilers can be used as heating options, but do not need to be used simultaneously. Therefore, the scenarios that can be selected for building energy system optimization include electric cooling and heating modes, conventional electric cooling and heating modes plus photovoltaic generation, etc.

We utilized Latin Hypercube Sampling (LHS) and Monte Carlo sampling methods to generate 16,320 samples by sampling the input parameters. This sampling method chosen for this study is common in building performance, analysis and can fully capture uncertainty characteristics and accelerate convergence [43]. This study utilized an Intel Core i7-8700 processor and Python 3.7.0 for simulation, prediction, and optimization. The environment was configured with the following versions of relevant libraries: pandas==0.23.4, gurobipy==9.1.0, numpy==1.19.5, scikit-learn==0.24.0, pytorch==1.0.2, torchvision==1.10.0, matplotlib==3.5.1, tqdm==4.64.0, sympy==1.8 and scipy==1.6.2.

3.2. Key features selection and data-enhancement

The model's two objective functions are divided into two parts. Specifically, part of Objective Function 1 is the carbon emission of the building body, and part is the carbon emission of the energy system. Similarly, Objective Function 2 represents both the building's and the energy system's cost. Due to the presence of excessive redundant data in construction cost and carbon emissions, it is challenging to obtain data using traditional optimization methods. Therefore, this study utilizes a GAN for data generation. Based on GAN feature extraction, building cost and carbon emission can be predicted. We comprehensively evaluate the influence of each input feature on LCCE and LCC, and select the key features. Of the input parameters, penetration rate, wall U value and cooling setpoint are the most significant features. Their influence on LCCE and LCC is 31.9 % and 26.3 % respectively. Another factor that influences LCCE is the heating setpoint. Due to the relatively low

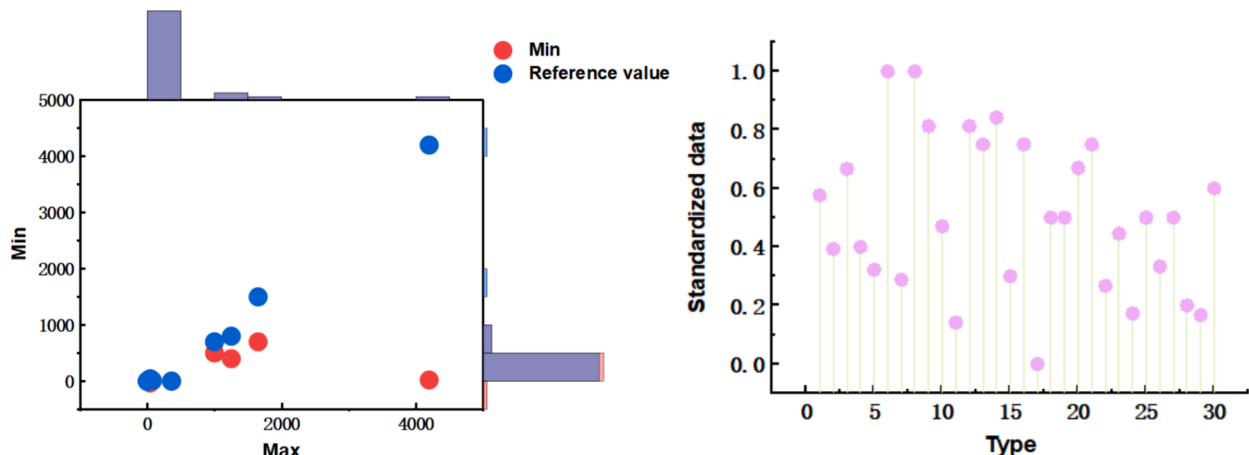


Fig. 7. Parameter table.

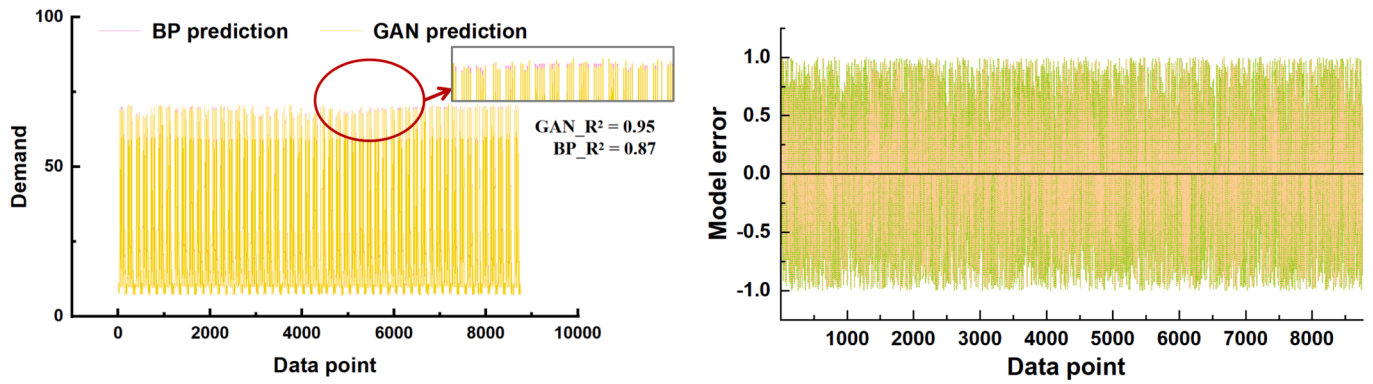


Fig. 8. Comparison of electric load prediction results and model error.

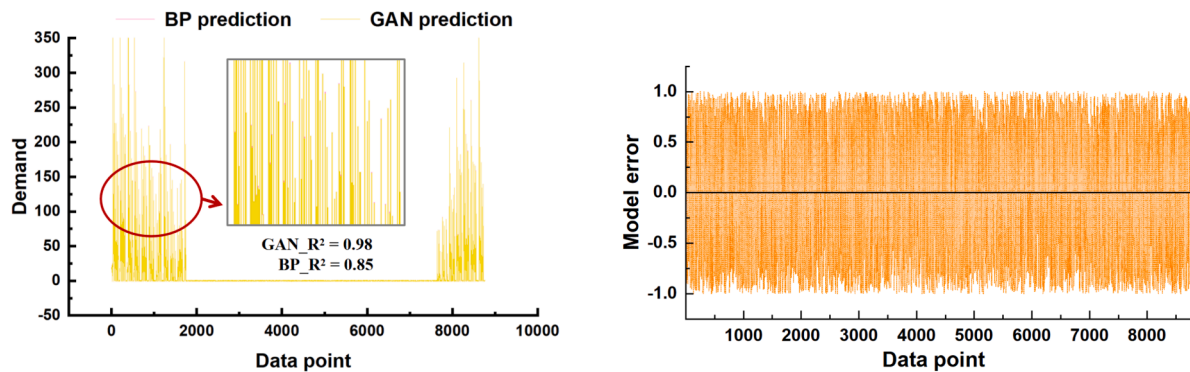


Fig. 9. Comparison of heat load prediction results and model error.

temperature in the office building, the effect of heating set point on LCCE was 22.5 %. Features with low contribution rates are similar across different outputs. For instance, the influence of fan efficiency and fan motor efficiency on all outputs is approximately 0.1 %, indicating that they are low impact factors in this study. This study considers the combined impact of each parameter on the output by averaging the contribution rate of both outputs to each parameter. Finally, the correlation degree of each feature was sorted in descending order, and the top 5 % of parameters with the highest correlation were selected as input features. On the other hand, unimportant parameters are eliminated during iteration. This process can eliminate unimportant parameters, reducing model complexity and improving operational speed. Fig. 7 illustrates the adoption dataset for the sampling method under various search methods. The R^2 value of prediction accuracy remains between 0.88 and 0.89, indicating the best prediction performance and the

shortest search time. The experimental calculation results indicate that the input features can generally be obtained from the 30 kinds listed, and that the descending order of correlation degree in the design phase is as follows: heating temperature in winter, wall heat absorption, U value of the wall, U value of the roof, cooling temperature in summer, building orientation, window to wall ratio(WWR), wall density, window solar heat gain absorption, U value of the window, rooftop solar absorption, number of energy users, coil heating efficiency, fresh air volume, illumination intensity, and wall solar energy absorption.

After data enhancement optimization, the R^2 value of the test dataset was greater than 0.95, while the R^2 value of the training dataset was above 95 %, meeting the requirements for perfect forecasting. This method performed 13.79 % better than the base neural network model. These two R^2 values have high accuracy, proving the applicability of the prediction model. To demonstrate accuracy, we randomly draw 8760

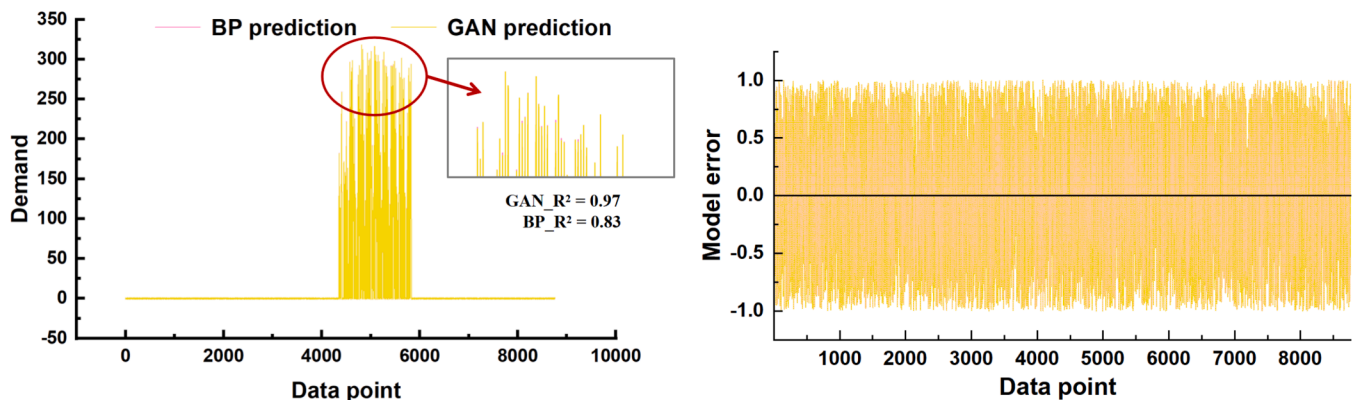


Fig. 10. Comparison of cooling load prediction results and model error.

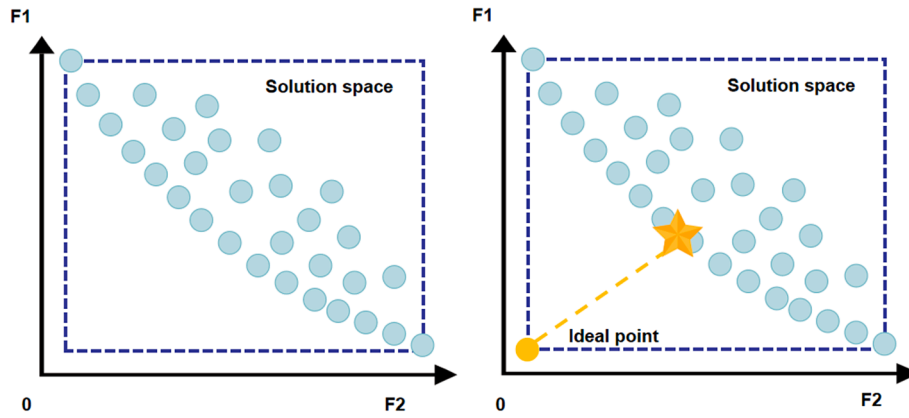


Fig. 11. Comparison of multi-objective optimization result.

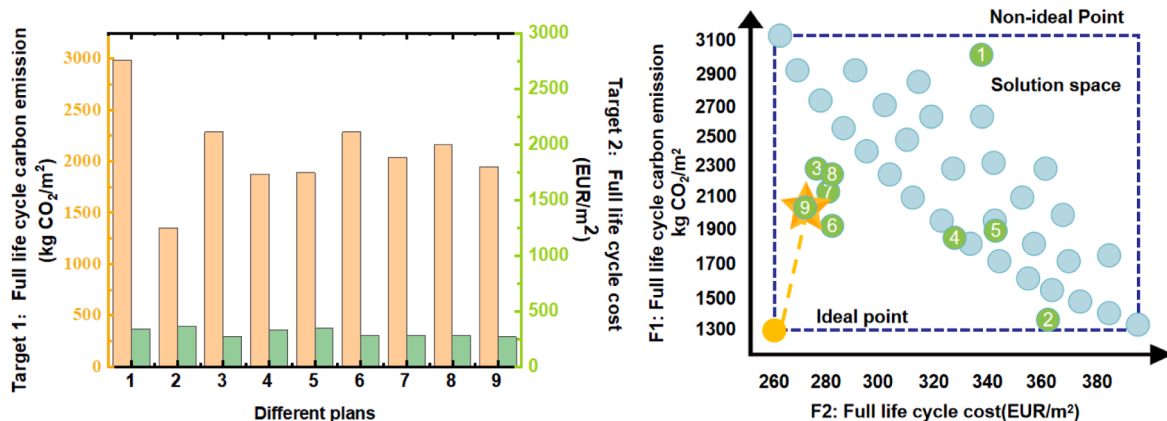
samples from the test set and the training set respectively. And analyzing the differences between actual results and simulation results for the test and training sets. If the prediction scale is one hour, the forecast situation of electrical load, thermal load and cooling load for 8760 h in a year is presented as the Figs. 8-10 below. After analyzing and comparing the BP prediction model and GAN prediction model, it can be seen that for electric load prediction, the R^2 value of the BP model and GAN model is 0.87 and 0.95, respectively, and the prediction accuracy is improved by 9.20 %. The R^2 value of the heat load BP model is 0.85, while the R^2 value of the GAN model is 0.98. This indicates that the GAN model has a 15.29 % higher prediction accuracy. At the same time, the R^2 value of the cold load BP model is 0.83, and the R^2 value of the GAN model is 0.97, indicating an increase in prediction accuracy of 16.87 %. The average increase of R^2 across all data types is 13.79 %, representing a significant improvement. The GAN model exhibits strong data supplementation performance in the absence of data.

It is worth noting that based on the approved method, not only can the values of LCCE and LCC be predicted simultaneously, but also the high prediction quality is maintained. There is good agreement between predicted and actual values for both the test and training sets. The prediction error is within an acceptable range, indicating that the model can be used as an alternative.

3.3. Multi-objective solution results

As the basis for multi-objective analysis, the multi-objective Pareto solution is illustrated in the Fig. 11. below. The main principle is to calculate the distance between each non-inferior solution and the optimal solution of each optimization target in the solution space, as shown on the left of the figure below. Next, sum the distances between the non-inferior solutions and the optimal solutions of different optimization targets. As shown on the right of the figure below, the star data with the shortest distance is the optimal solution.

The multi-objective results indicate that the optimization results of data enhancement using AHA-VaEA-NSGA-III are superior to those of other algorithms. The integration of algorithms can improve prediction accuracy, while increasing the time consumption of each prediction. However, compared to the simulation time for each dataset in EnergyPlus, which is approximately 50 h, the prediction and optimization speed of this research strategy can significantly improve efficiency. Increasing computational speed can enhance study practicality and enable the solution of more complex problems. Otherwise, the LCCE ranges from 8295.12 to 18972.84 kg CO₂/m², which is smaller than that of other algorithms that range from 20320.63 to 36408.51 kg CO₂/m². On the other hand, the LCC of data enhancement multi improved NSGA-III can reach about 358.28 EUR/m², which is smaller than the best value of algorithms. It indicates that the data enhancement improved NSGA-III



Note: 1 represents base scenario, 2 represents optimal carbon emission plan, 3 represents optimal cost plan, 4 represents simple target addition, 5 represents NSGA-III algorithm, 6 represents AHA-NSGA-III algorithm, 7 represents VaEA-NSGA-III algorithm, 8 represents AHA-VaEA-NSGA-III algorithm, 9 represents GAN-AHA-VaEA-NSGA-III algorithm.

Fig. 12. Comparison of multi-objective optimization result. Note: 1 represents base scenario, 2 represents optimal carbon emission plan, 3 represents optimal cost plan, 4 represents simple target addition, 5 represents NSGA-III algorithm, 6 represents AHA-NSGA-III algorithm, 7 represents VaEA-NSGA-III algorithm, 8 represents AHA-VaEA-NSGA-III algorithm, 9 represents GAN-AHA-VaEA-NSGA-III algorithm.

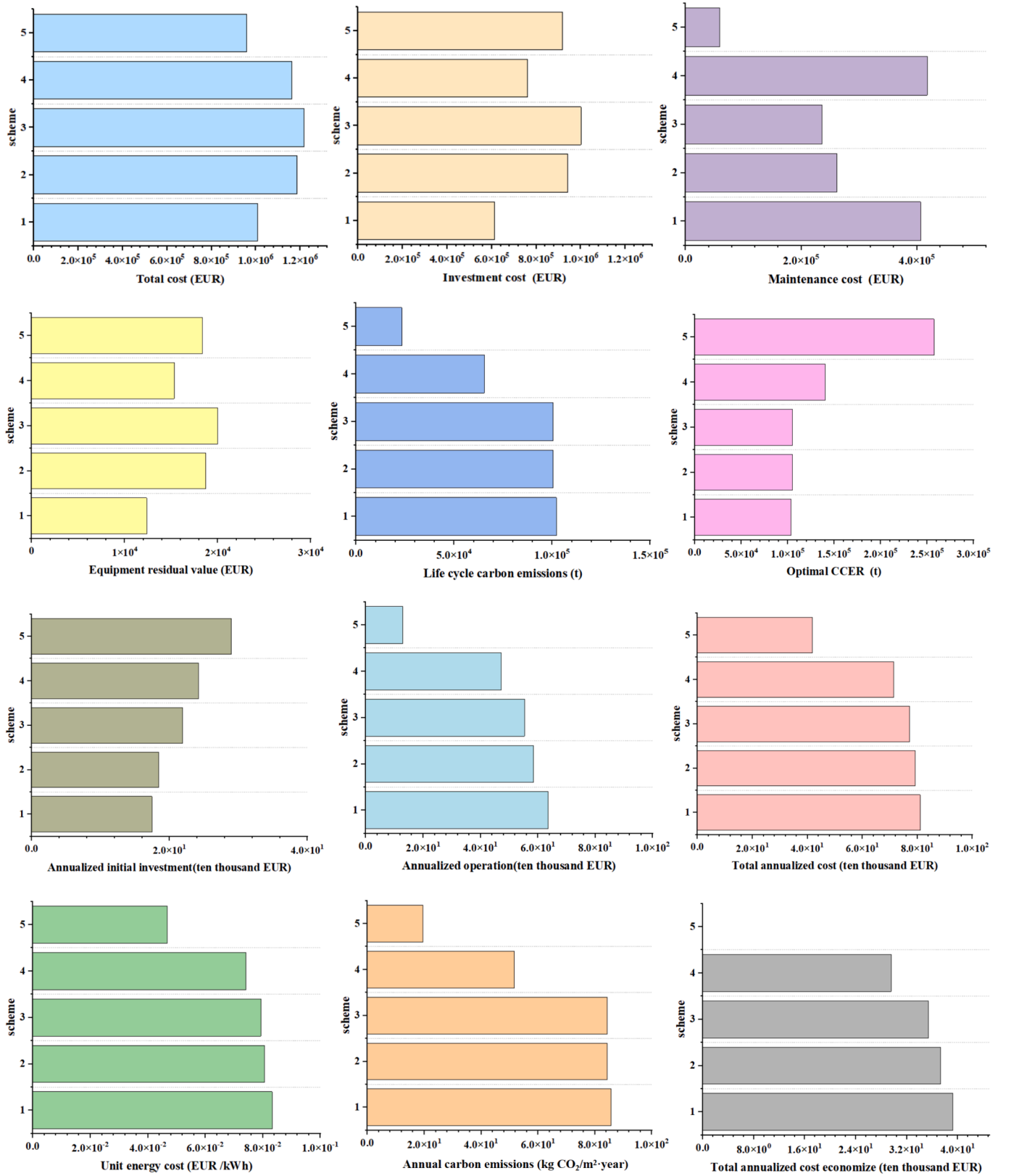


Fig. 13. Technical, economic, and environment analysis comparison of different schemes.

is more concentrated and does not fully converge. The distribution of Pareto points in data enhancement multi improved NSGA-III is relatively uniform and distributed along the coordinate axis. After selecting the best optimization algorithm, we examined various schemes among the optimal solutions analyzed by the algorithm, considering their

suitability for this design.

This paper analyzes nine scenarios, with details available in Fig. 12. The optimal solution can be obtained at a cost of 268.20 EUR/m², and carbon emission of 1946.33 kgCO₂/m². Compared to the baseline scenario, the simple target plan showed a 37.22 % decrease in full life cycle

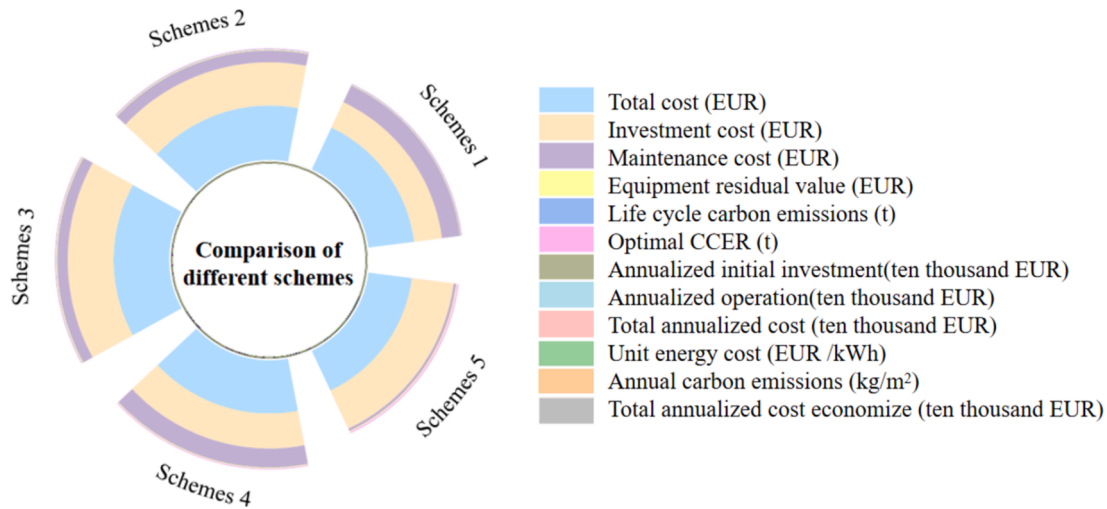


Fig. 14. Comprehensive comparison of different schemes.

carbon emissions (target 1) and a 1.31 % increase in full life cycle costs (target 2). The cost-optimized NSGA-III plan resulted in a 36.68 % decrease in full life cycle carbon emissions (target 1) and a 1.89 % increase in full life cycle costs (target 2). For the improved method scenario, compared to the baseline, the full life cycle carbon emission (target 1) of the AHA-NSGA-III plan decreased by 23.41 %, and the full life cycle cost (target 2) decreased by 16.65 %; the full life cycle carbon emission (target 1) of the VaEA-NSGA-III plan decreased by 31.62 %, and the full life cycle cost (target 2) decreased by 10.43 %; the full life cycle carbon emission (target 1) of the AHA-VaEA-NSGA-III plan decreased by 27.51 %, and the full life cycle cost (target 2) decreased by 13.58 %; the full life cycle carbon emission (target 1) of the GAN-AHA-VaEA-NSGA-III plan decreased by 34.74 %, and the full life cycle cost (target 2) decreased by 20.69 %. It is worth noting that Plan 9, named GAN-AHA-VaEA-NSGA-III, is the most balanced among all schemes. The improved NSGA-III algorithm not only considers the balance of optimization effects but also maximizes the joint optimization effects. Compared to other methods, the optimization method is continuously improved and is more practical and reasonable than the traditional method. It can also be extended to other studies.

4. Analysis and verification

To verify the feasibility of the model under different energy forms, the study takes different sustainable energy system plan schemes as examples. The analysis further examines the investment cost, operation and maintenance cost, equipment salvage value, and total cost under different investment plans. This study compares five schemes.

- Scheme 1: the traditional electric cooling and heating mode, which comprises the power grid, electric boiler, and water chilling unit.
- Scheme 2: the conventional electric cooling and heating mode, as well as photovoltaic generation.
- Scheme 3: traditional electric cooling and heating mode, and photovoltaic generation, and energy storage.
- Scheme 4: traditional electric cooling and heating mode, and photovoltaic generation, and energy storage, and pump.
- Scheme 5: traditional electric cooling and heating mode, and photovoltaic generation, and energy storage, and pump, and hydrogen.

To obtain multi-objective optimization results, the optimal solution should be derived from the Pareto optimal solution set. This dataset comprises decision variable solutions that satisfy the Pareto optimal state. Because the optimization objectives are contradictory, the results

of each multi-objective algorithm are expected to show a convergence trend. The traditional scheme is often to explore the best spot from the Pareto frontier [44]. Once the ideal point is determined, the distance between each optimal point in the Pareto front set and the ideal point is calculated. However, because of the different ranges of changes for each target, some targets with large changes will affect site selection and the final scheme. This study adopts the multi improved NSGA-III method, as described above, for scheme selection to reduce the influence of different variation ranges on scheme selection. Technoeconomic assessments of different schemes can be visually compared in Fig. 13.

The results show that the value of LCCE fluctuated between 8295.08 kg CO₂/m² and 36408.52 kg CO₂/m². The 5 schemes' mean value of LCC fluctuated between 342.10 EUR/m² and 434.04 EUR/m². Regarding annual carbon emissions under different investment scenarios, the carbon emission of scheme 1 is 85.72 kg CO₂/m²·year. The annual carbon emission of scheme 2 is 84.36 kg CO₂/m²·year. The annual carbon emission of scheme 3 is 84.36 kg CO₂/m²·year, and it is the same as that of scheme 2. Because PV is not selected compared to storage, which only changes the carbon emissions timeline, but does not reduce it. The scheme 5 has the lowest annual carbon emissions and relatively low total cost. Therefore, in the design phase, the life cycle carbon emission optimization approach for scheme 5 can strike a balance between cost and carbon emissions, which are 54.78 tCO₂e/year and account for nearly a quarter of pure power systems. At the same time, the optimal carbon emission reduction of scheme 5 can reach 257337.62 t, which is the highest among the five schemes.

Based on the above analysis, a comprehensive comparison of the different schemes is presented in Fig. 14. One can readily ascertain that the best plan is scheme 5, which incorporates both electric cooling and heating, photovoltaics, energy storage, pumps, and hydrogen.

5. Conclusion

This study proposes a new strategy that integrates building and energy systems to obtain the system design scheme from a whole-system full life cycle perspective. The results prove that this strategy is an effective method to balance LCCE and LCC, and makes the optimization results of each objective more reasonable, outperforming traditional methods. The main conclusions of this study are as follows:

- Through the integration of input feature evaluation and selection, high prediction accuracy can be achieved. The optimization performance and convergence degree of the multi-optimized NSGA-III (GAN-AHA-VaEA-NSGA-III) algorithm are higher than those of other

algorithms, and the average R^2 value of GAN performs 13.79 % better than the baseline neural network model.

- The proposed method in this paper considers the optimization of two core objectives, effectively addressing the issue of differing optimization ranges for each objective and minimizing the overall optimization value. LCCE is reduced by 34.74 %, and LCC is reduced by 20.69 %, both representing improvements over the traditional method and potentially extendable.
- This strategy can effectively optimize the whole-energy system performance from a life-cycle perspective, and we significantly improve the optimization effect and calculation speed compared to the traditional method.

As a problem-oriented research, future studies can broaden the strategy and investigate other more detailed issues in the preliminary design or running stage of buildings. This study proposes a novel strategy that considers the input and variation range of each objective to improve the optimization result and better balance the dual optimization objective using the proposed multi-improved NSGA-III algorithm even without historical data. The model is practical and universal, and can effectively reduce the impact of different target ranges, and can be widely used in engineering activities.

CRediT authorship contribution statement

Ying Tian: Writing – original draft, Validation, Conceptualization.
Kang Chai: Writing – review & editing, Methodology, Funding acquisition, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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